

Breast Tumor Classification: Decision Tree vs. AdaBoost

Comparative analysis of tree-based classifiers | Farhang Khanjani

1. Objective

The goal of this project is to build and compare classification models that distinguish malignant (M) from benign (B) breast tumors based on 30 numeric features computed from cell-nucleus measurements. Two model families are evaluated: a single Decision Tree with tuned depth and splitting criterion, and an AdaBoost ensemble of decision stumps. The comparison covers predictive accuracy, bias–variance behavior, and decision-boundary structure.

2. Data & Methodology

The dataset contains tumor records with an ID, a diagnosis label (M/B), and 30 continuous features. Labels were encoded (M→1, B→0) and the data was split into 75% training and 25% testing sets with stratification to preserve class balance (random_state=50 for reproducibility).

Model selection was performed via systematic hyperparameter sweeps: Decision Trees across three splitting criteria (gini, entropy, log_loss) and depths 1–3; AdaBoost across 300–500 estimators (learning rate 0.5). For visualization, the feature space was reduced to two dimensions with PCA and both best models were retrained to plot their decision boundaries on test data.

3. Decision Tree Results

Criterion	Max Depth	Train Accuracy	Test Accuracy
gini	1	0.9178	0.9371
gini	2	0.9484	0.9580
gini	3	0.9695	0.9720

entropy	1	0.9108	0.9231
entropy	2	0.9178	0.8811
entropy	3	0.9624	0.9301
log_loss	1	0.9108	0.9231
log_loss	2	0.9178	0.8811
log_loss	3	0.9624	0.9301

The best single tree uses the **gini** criterion at **depth 3**, reaching **97.2% test accuracy**. Accuracy improves consistently with depth, and gini outperforms entropy/log_loss at every depth on this dataset.

4. AdaBoost Results

Base Depth	N Estimators	Train Accuracy	Test Accuracy
1	300	1.0000	0.9860
1	400	1.0000	0.9860
1	500	1.0000	0.9860
2	300	1.0000	0.9860
2	400	1.0000	0.9860
2	500	1.0000	0.9860

AdaBoost reaches **98.6% test accuracy**, stable across all estimator counts — performance saturates by 300 estimators, so the smallest ensemble is preferred for efficiency.

5. Model Comparison

Model	Configuration	Test Accuracy	Bias–Variance Profile
Decision Tree	gini, depth 3	0.9720	Low bias, moderate variance
AdaBoost	300 stumps, lr 0.5	0.9860	Low bias, low variance

The PCA-based decision-boundary visualization shows the single tree carving axis-aligned rectangular regions, while AdaBoost forms a finer, better-adapted boundary. This is consistent with the accuracy gap: boosting many weak learners reduces variance without sacrificing bias, cutting the test error roughly in half relative to the best single tree (1.4% vs. 2.8%).

6. Conclusion

AdaBoost with 300 depth-1 estimators is the recommended model, achieving 98.6% test accuracy with a favorable bias–variance profile. The experiment illustrates a core practical lesson: a well-tuned ensemble of weak learners outperforms a deeper standalone tree while remaining resistant to overfitting. All experiments are fully reproducible (fixed random seeds, stratified splits) and implemented in Python with scikit-learn.

Full source code and outputs are available in the accompanying Jupyter notebook.